

THE MONTREAL FUNGUIDE

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The Montreal FunGuide is a RESTful web service that provides a plentitude of information regarding various fungi species you can find right here in your backyard. This service also includes fun things to do like recipes you can cook or trips you can take to find fungi all over the city. There is a map included in this service with locations thanks to the abundance of data received from our Composite APIs.

Tables in Descending Order

Table Name	Fields
users	7
registered_device	6
fungi	8
species	10
habitat	6
recipe	8
ingredient	6
calorie	5
map	4
location	9
trip	7
trip_location	5
fungi_location	5
fungi_recipe	5

List of Tables in Depth:

Table Name	Description	Feilds
users	Stores user accounts for authentication and authorization.	7
registered_device	Tracks devices linked to user accounts for security monitoring.	6
fungi	Raw fungi dataset imported from GBIF.	8
species	Curated fungi species data from GBIF.	10
habitat	Environmental zones in Montreal where fungi are found.	6
recipe	Recipes retrieved from external APIs.	8
ingredient	Ingredients used in recipes (Ifungi).	6
calorie	Nutritional info for ingredients & fungi.	5
map	Container for mapping system.	4
location	Geographic coordinates of fungi in MTL.	9
trip	User-created exploration routes.	7
trip_location	Junction table linking trips to ordered locations.	5
fungi_location	Junction table linking species to locations.	5
fungi_recipe	Junction table linking species to recipes.	5

* Note that API resources are subject to change as the project is completed.

API resources:

GBIF	https://www.gbif.org/
Air Quality API	https://api-ninjas.com/api/airquality
Weather API	https://api-ninjas.com/api/weather#weather-endpoint
Nutrition API	https://api-ninjas.com/api/nutrition#nutrition-endpoint
Recipe API	https://api-ninjas.com/api/recipe
Geocoding API	https://api-ninjas.com/api/geocoding

This web service introduces 4 main groups of information.

This includes resources for Authorization, Fungi information, Location information and Recipe information.

AUTH GROUP

Table Name	Collection Route
users	/auth/users
registered_device	/auth/users/{user_id}/devices

FUNGI GROUP

Table Name	Collection Route
fungi	/fungi/{species_id}/{fungi_id}
species	/fungi/species
habitat	/fungi/habitats
fungi_location	/fungi/{species_id}/{fungi_id}/{location_id}

LOCATION GROUP

Table Name	Collection Route
map	/map
location	/map/locations
trip	/map/trips
trip_location	/map/trips/{trip_id}/locations

RECIPE GROUP

Table Name	Collection Route
recipe	/recipes
ingredient	/recipes/{recipe_id}/ingredients
calorie	/recipes/{recipe_id}/ingredients/{ingredient_id}/calories
fungi_recipe	/fungi/{species_id}/recipes

Table Filtering

Field Name	Data Type	Filter Parameter	HTTP Method	Example
family	VARCHAR	fungi-family	GET	/species?fungi-family=agaricus
is_edible	BOOLEAN	is-edible	GET	/species?is-edible=1
toxicity_level	ENUM	toxicity	GET	/species?toxicity=deadly
habitat	VARCHAR	habitat	GET	/species/{id}/habitats?habitat=forest
discovery_year	INT	discovery-year	GET	/species?discovery-year=2000
difficulty	ENUM	difficulty	GET	/recipes?difficulty=easy
calories	INT	calories-max	GET	/recipes?calories-max=333
ingredient	VARCHAR	ingredient	GET	/recipes/{id}/ingredients?ingredient=mushroom
borough	VARCHAR	borough	GET	/map?borough=Verdun
trip_distance	FLOAT	distance-max	GET	/map/trips?distance-max=10

Sorting Parameter

Resource	Sortable Field	Example
species	family	/species?sort=family
species	discovery_year	/species?sort=discovery_year.desc
recipes	calories	/recipes?sort=calories
recipes	difficulty	/recipes?sort=difficulty
location	borough	/map?sort=borough
trip	trip_distance	/map/trips?sort=trip_distance

Pagination

Resource	Type	Parameter	Example
species	offset	limit, offset	/species?limit=20&offset=0
recipes	offset	limit, offset	/recipes?limit=10&offset=0
map	offset	limit, offset	/map?limit=15&offset=0
trip	offset	limit, offset	/map/trips?limit=5&offset=0

